COS326 Semester Test 2 2026

Final Grade

61.5 / 90

Your instructor added question feedback to this assessment

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61.5 / 90

ASSESSMENT CONTENT

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1 **MULTIPLE CHOICE** CORRECT 1 / 1

A university is redesigning its student management system. The development team has created a relational schema in which students, module registrations, and modules are stored in separate tables. The application uses joins to retrieve registration information and then reconstructs student objects containing collections of registered modules. Although the system maintains data consistency, the developers argue that the database design does not naturally reflect the structure of the objects used in the application. Which explanation best justifies adopting an object-oriented database for this system?

- A. **Correct:** Relational databases represent data using relations, whereas object-oriented databases can represent objects with identity, references, and complex nested structures. — Correct answer
- B. Relational databases execute joins by retrieving complete tables into application memory before matching related records.
- C. Relational databases cannot represent one-to-many relationships without duplicating student records for every registered module.
- D. Relational databases require every relationship between students and modules to be implemented outside the database management system.

2 MULTIPLE CHOICE CORRECT 1 / 1

A laboratory application stores `Equipment` objects in ObjectDB. Two equipment objects have identical values for `name`, `category` and `location`, but they represent two different physical devices. An administrator later changes the location of one device. Which design principle is most important for distinguishing the two stored objects?

- A. Require every object to have a different value for every attribute.
- B. **Correct:** Give each independent object a persistent identity, such as an OID. — Correct answer
- C. Use the object's current location as its permanent identifier.
- D. Store each object as a separate XML document with a unique root tag.

3 MULTIPLE CHOICE CORRECT 1 / 1

A developer uses ObjectDB with JPA to retrieve equipment records. The following query executes successfully:

```
SELECT e FROM Equipment e
WHERE e.replacementCost > 10000
```

The developer later renames the database's underlying storage structure without changing the `Equipment` entity or its persistent attributes. The query continues to work. Which property of JPQL best explains this behaviour?

- A. **Correct:** JPQL uses the entity model as its query abstraction, reducing dependence on the database's physical storage representation. — Correct answer
- B. JPQL translates entity attributes into fixed database column names during compilation, independently of the persistence provider.
- C. JPQL automatically detects changes to entity attributes and rewrites the Java source code before execution.
- D. JPQL queries are evaluated exclusively against Java objects currently held in application memory.

4 **MULTIPLE CHOICE** CORRECT 1 / 1

A logistics company uses PostgreSQL to manage equipment maintenance. Developers need to introduce a custom `GeoLocation` data type and store multiple inspection readings for each piece of equipment. The existing application relies on SQL queries, relational constraints, and transactions, which must remain supported.
Which database design decision most directly satisfies these requirements without abandoning the relational model?

- A. Replace the relational schema with XML documents and implement inspection retrieval using XPath expressions.
- B. **Correct:** Extend the relational schema with user-defined types and collection-supporting features provided by PostgreSQL. Correct answer
- C. Map each equipment record to a Java object and persist its attributes exclusively through application-level object serialisation.
- D. Introduce a separate graph database and represent equipment, locations, and inspections as interconnected nodes.

5 **MULTIPLE CHOICE** CORRECT 1 / 1

A vehicle-inspection system uses PostgreSQL. The developer defines a composite type and a table as follows:

```
CREATE TYPE VehicleInfo AS (
    registration_no VARCHAR(15),
    make VARCHAR(30),
    inspection_result VARCHAR(20)
);

CREATE TABLE Inspection (
    inspection_id SERIAL PRIMARY KEY,
    vehicle VehicleInfo
);
```

A developer argues that the `vehicle` column violates the relational approach because it groups several attributes into one column.

- A. The design is invalid because every column in a relational table must contain a primitive data type.

B. The design converts PostgreSQL into a purely object-oriented DBMS because composite types eliminate relational tables.

C. The design is valid only if each attribute of VehicleInfo is stored in a separate table linked by foreign keys.

D. **Correct:** The design is object-relational because a user-defined composite type allows structured attributes to be stored within a relational table. Correct answer

6 **MULTIPLE CHOICE** CORRECT 1 / 1

A university stores the weekly study hours reported by each student in a PostgreSQL array. The table is defined as follows:

```
CREATE TABLE StudentStudy (
   student_id INT PRIMARY KEY,
   weekly_hours INT[]
);
```

A developer executes the following statement:

```
INSERT INTO StudentStudy
VALUES (101, ARRAY[12, 15, 10, 18]);
```

The developer then executes:

```
SELECT weekly_hours[2:3]
FROM StudentStudy
WHERE student_id = 101;
```

What does the query return?

A. An array containing `12`, `15` and `10`, because array slices start at index zero.

B. **Correct:** An array containing `15` and `10`, because the slice includes both specified positions. Correct answer

C. A single integer, `15`, because PostgreSQL arrays use zero-based indexing.

D. An error, because PostgreSQL does not support retrieving a range of array elements using subscripting.

7 **MULTIPLE CHOICE** CORRECT 1 / 1

What is the data type of an inventoryRecord defined as follows in PostgreSQL?

```
CREATE TYPE inventoryRecord AS (
  item_name text,
  quantity integer
);
```

- A. **Correct:** Composite — Correct answer
- B. Enumerated
- C. Text
- D. Integer

8 **MULTIPLE CHOICE** CORRECT 1 / 1

A research group receives XML files from several institutions. Each file contains researcher information, but one institution includes an optional `ORCID` element while another includes an optional `ResearchGroup` element. The files still follow a recognisable hierarchical structure.
Why is semi-structured data a suitable way to describe these files?

- A. **Correct:** It supports organised data whose structure can vary between records. — Correct answer
- B. It requires all information to be stored in fixed-width relational columns.
- C. Semi-structured data prohibits optional attributes and elements.
- D. It prevents applications from identifying relationships between elements.

9 **MULTIPLE CHOICE** CORRECT Full grades given 1 / 1

A hospital stores patient admission records in `admissions.xml`, which is queried using BaseX. The document contains multiple `WARD` elements, each containing several `PATIENT` elements. Each patient has an `ADMISSION` element with an `ADMISSIONNO`

child.
Wandile, a database developer needs an XPath expression that returns the admission number of the first patient in the second ward, regardless of how many admissions that patient has.
Which expression correctly selects the required node for Wandile?

- A. `(//WARD/PATIENT/ADMISSION/ADMISSIONNO)[2]`
- B. `(//WARD[2]/PATIENT[1]/ADMISSION/ADMISSIONNO)[1]` — Correct answer
- C. **Incorrect:** `//WARD[2]/PATIENT[1]//ADMISSIONNO`
- D. `//WARD/PATIENT[1]/ADMISSION[2]/ADMISSIONNO`

10 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

An HR department stores employee data in `employees.xml`. The following XQuery should calculate the average salary for each branch:

```
for $b in distinct-values(//employee/branch)
let $s := //employee[branch = $b]/salary
return
   <branch name="{$b}">{avg($s)}</branch>
```

The query returns one result for each distinct branch name. However, the XML document contains employees from different regional offices that share the same branch name.
Which change best ensures that the averages are calculated for the intended branch elements?

- A. Bind `$b` to each branch element and select its employee salaries relative to that element. — Correct answer
- B. Replace `avg($s)` with `sum($s) div count($s)`.
- C. **Incorrect:** Apply `distinct-values()` to `$s` before calculating the average.
- D. Replace `distinct-values()` with `count()`.

11 **MULTIPLE CHOICE** CORRECT 1 / 1

A XQuery expression iterates over employee elements using a `for` clause. Thandi, an HR manager now wants the result to include only employees whose salary exceeds 40,000, while preserving the existing iteration and return structure.
Which change is most appropriate for Thandi to make?

- A. Use `let` to delete employees below the threshold.
- B. **Correct:** Add the salary condition to a `where` clause. Correct answer
- C. Replace the `return` clause with `order by`.
- D. Replace the `for` clause with an XML namespace declaration.

12 **MULTIPLE CHOICE** INCORRECT 0 / 1

Jaco, a researcher stores the following document in BaseX as `research.xml`:

```
<researchers>
 <researcher id="R1">
   <name>Ada</name>
   <projects>
     <project><title>Forensics</title></project>
     <project><title>Blockchain</title></project>
   </projects>
 </researcher>
 <researcher id="R2">
   <name>Ben</name>
   <projects>
     <project><title>Forensics</title></project>
   </projects>
 </researcher>
</researchers>
```

Jaco, then executes this XQuery:

```
for $r in //researcher
where $r/projects/project/title = "Forensics"
return count($r//project)
```

What sequence of values does BaseX return?

A. `2, 1` Correct answer

B. **Incorrect:** `1, 2`

C. `2, 2`

D. `1, 1`

13 **MULTIPLE CHOICE** **CORRECT** Full grades given **1 / 1**

A university stores research project records in BaseX. Each `PROJECT` contains a `STATUS` element and a `SUPERVISOR` element. Takalani, researcher wants to retrieve the supervisor of the second project whose status is `Approved`, ignoring projects with any other status.

The relevant XML structure is:

```
<PROJECTS>
   <PROJECT>
       <STATUS>Approved</STATUS>
       <SUPERVISOR>Dr Du Toit</SUPERVISOR>
   </PROJECT>
   <PROJECT>
       <STATUS>Pending</STATUS>
       <SUPERVISOR>Dr Naidoo</SUPERVISOR>
   </PROJECT>
   <PROJECT>
       <STATUS>Approved</STATUS>
       <SUPERVISOR>Prof Dube</SUPERVISOR>
   </PROJECT>
   <PROJECT>
       <STATUS>Approved</STATUS>
       <SUPERVISOR>Dr Smith</SUPERVISOR>
   </PROJECT>
</PROJECTS>
```

Which XPath expression returns the required supervisor?

A. `(/PROJECTS/PROJECT[STATUS='Approved']/SUPERVISOR)[2]`

B. `/PROJECTS/PROJECT/SUPERVISOR[2][../STATUS='Approved']`

C. **Correct:** `(/PROJECTS/PROJECT[STATUS='Approved'])` Correct answer

```
[2]/SUPERVISOR
```

D. `/PROJECTS/PROJECT[2][STATUS='Approved']/SUPERVISOR`

14 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

Thabo Mokoena, a data analyst at a South African transport company, uses MongoDB to analyse vehicle telemetry records. Each document contains a vehicle's speed, its location and an array of recorded alerts.
An example document is shown below:

```
{
   vehicleId: "BUS104",
   speed: 85,
   location: {
       zone: "North",
       road: "R21"
   },
   alerts: ["overspeed", "engine_warning"]
}
```

Thabo must retrieve all vehicles that are in the North zone and have an `overspeed` alert, regardless of whether other alerts are also present.
Which MongoDB query should he execute?

A.
```
db.telemetry.find({
   "location.zone": "North",
   alerts: { $all: ["overspeed", "engine_warning"] }
})
```

B.
```
db.telemetry.find({
   "location.zone": "North",
   alerts: "overspeed"
})
```
Correct answer

Incorrect:
```
db.telemetry.find({
   "location.zone": "North",
   alerts: ["overspeed"]
})
```

D.
```
db.telemetry.find({
   location: "North",
   alerts: { $in: ["engine_warning"] }
})
```

15 **MULTIPLE CHOICE** CORRECT 1 / 1

An online banking platform replicates customer account data across three data centres. During a network partition, one data centre cannot communicate with the other two. The isolated data centre continues accepting balance enquiries and returns its locally stored values, while the other data centres process recent transactions.
Which statement best characterises the system's behaviour during the partition?

- A. **Correct:** The system favours availability over strong consistency while tolerating the partition, allowing responses that may not reflect the latest transactions. — Correct answer
- B. The system sacrifices partition tolerance to preserve availability, since the isolated data centre continues processing requests.
- C. The system preserves strong consistency because each data centre responds using a valid local copy, even if replicas differ.
- D. The system guarantees consistency and availability because replication ensures that every data centre eventually receives the same updates.

16 **MULTIPLE CHOICE** CORRECT Full grades given 1 / 1

A Security Operations Centre stores security events in MongoDB. Analyst Naledi needs to retrieve events that are either high severity (4 or above and still open) or critical severity (5) regardless of status.
The collection contains the following documents:

```
{ eventId: 201, severity: 2, status: "Open" }
{ eventId: 202, severity: 4, status: "Closed" }
{ eventId: 203, severity: 5, status: "Open" }
{ eventId: 204, severity: 3, status: "Closed" }
{ eventId: 205, severity: 4, status: "Open" }
{ eventId: 206, severity: 5, status: "Closed" }
```

Which query correctly implements Naledi's requirements?

- A.

```
db.events.find({
   $or: [
       { severity: { $gte: 4 } },
       { status: "Open" }
   ]
```

```
})

db.events.find({
   severity: { $gte: 4 },
   status: "Open"
})

db.events.find({
   $and: [
       { severity: { $gte: 4 } },
       { $or: [{ status: "Open" }, { severity: 5 }] }
   ]
})
```

B.

C.

Correct:

```
db.events.find({
   $or: [
       { severity: { $gte: 4 }, status: "Open"
},
       { severity: 5 }
   ]
})
```

D. Correct answer

17 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

Isiqalo, an e-commerce company stores customer profiles in MongoDB. Its collection contains documents created by different versions of the application:

```
// Older application version
{
   customerId: 101,
   name: "Thandi Nkosi",
   email: "thandi@isiqalo.com"
}

// Newer application version
{
   customerId: 102,
   name: "James Smith",
   email: "james@isiqalo.com",
   preferences: {
       language: "English",
       notifications: true
   }
}
```

The company deploys the newer application without migrating the older documents. Both document types remain in the same collection, and the application queries customers by `customerId`.
Which statement best explains the compatibility of these documents?

A. MongoDB stores each document using a separate implicit schema, preventing queries from matching documents created by different application versions.

B. MongoDB infers a collection-wide schema from the newest document and applies missing fields automatically to older documents.

C. **Correct:** MongoDB permits documents with different structures in one collection, although application logic or schema validation may still impose consistency requirements. — Correct answer

D. MongoDB automatically converts nested fields into top-level fields whenever a document omits attributes present in newer documents.

---

18 **MULTIPLE CHOICE** **INCORRECT** **0 / 1**

A mining company stores equipment maintenance records in MongoDB. Each record contains an array of embedded inspection entries.

```
{
   equipmentId: "EXC-408",
   inspections: [
       { result: "Pass", inspector: "Lerato" },
       { result: "Fail", inspector: "Amara" },
       { result: "Pass", inspector: "Pieter" }
   ]
}
```

The maintenance manager needs to retrieve equipment records where the second inspection was conducted by Amara. Records must not match merely because Amara appears elsewhere in the array.
Which query correctly implements this requirement?

A.
```
db.equipment.find({
    "inspections.inspector": "Amara"
})
```

B. **Incorrect:** `db.equipment.find({`
`"inspections.2.inspector": "Amara"`

```
})
```

C.
```
db.equipment.find({
    "inspections": { inspector: "Amara" }
})
```

D.
```
db.equipment.find({
    "inspections.1.inspector": "Amara"
})
```
Correct answer

19 **MULTIPLE CHOICE** CORRECT 1 / 1

An online publisher stores article engagement records in MongoDB. Each document records an article title and the number of likes received during a particular reporting period.

```
{ title: "Tech Today", likes: 180 }
{ title: "Tech Today", likes: 160 }
{ title: "Data Weekly", likes: 220 }
{ title: "Data Weekly", likes: 120 }
{ title: "Secure World", likes: 290 }
```

Ndingoho, an analyst must identify articles whose combined likes exceed 300. Ndingoho filters individual documents using `likes > 150` before calculating totals. This produces an incorrect report because some documents are excluded before aggregation.
Which change addresses the underlying error?

A. Replace `$sum` with `$avg` so excluded documents contribute to the total indirectly.

B. **Correct:** Apply the threshold to the grouped totals instead of filtering individual documents by `likes`.
Correct answer

C. Group by `likes` rather than `title` so that all engagement records are retained.

D. Sort the documents by `likes` before applying the original filter.

20 **MULTIPLE CHOICE** CORRECT 1 / 1

A security analyst wants to find documents in the `blogs` collection whose `comments` field contains the search term `"phishing"` using MongoDB's `$text` operator. No text index currently exists.

What should the analyst do first?

A. Create a graph relationship for each comment.

B. **Correct:** Create a text index on the `comments` field. Correct answer

C. Convert all comments into an integer array.

D. Create a unique index on the `_id` field.

21 **MULTIPLE CHOICE** **CORRECT** **1 / 1**

Kagiso executes an aggregation query in the MongoDB shell:

```
let res = db.incidents.aggregate([
   { $match: { status: "Open" } },
   { $project: { _id: 0, incidentId: 1, severity: 1 } }
]);
```

Kagiso wants to print every returned document exactly once. He writes the following loop:

```
while (res.hasNext()) {
   printjson(res.hasNext());
}
```

Assuming the cursor initially contains matching documents, what is the most accurate assessment of this loop?

A. **Correct:** It repeatedly prints `true` because `hasNext()` checks for a result without consuming it. Correct answer

B. It prints `false` after the first iteration because `hasNext()` invalidates the cursor.

C. It raises an error because aggregation cursors do not support `hasNext()`.

D. It prints each document because `hasNext()` advances the cursor after checking availability.

22 **MULTIPLE CHOICE** CORRECT 1 / 1

A cybercrime investigator models communication records in Neo4j. The graph contains the following relationships:

```
CREATE
 (l:Lebo {name: 'Lebo'}),
 (m:Person {name: 'Musa'}),
 (z:Person {name: 'Zanele'}),
 (l)-[:CONTACTED]->(m),
 (z)-[:CONTACTED]->(m);
```

The investigator considers the following proposed addition to the graph:

```
CREATE (m)-[:CONTACTED]->(l);
```

What would this addition change about the graph?

A. It would replace the existing relationship from Lebo to Musa with a relationship in the opposite direction.

B. It would create a second node representing Musa because each directed relationship requires a distinct endpoint.

C. **Correct:** It would establish reciprocal `CONTACTED` relationships between Lebo and Musa without changing Zanele's existing relationship. Correct answer

D. It would make all relationships involving Musa bidirectional, including the relationship from Zanele.

23 **MULTIPLE CHOICE** CORRECT 1 / 1

A university stores student information in Neo4j. During data migration, the following nodes are created:

```
CREATE
 (:Person {name: 'Amogelang', residence: 'Hatfield'}),
 (:Person {name: 'Ruan'}),
 (:Person {name: 'Jabulani', residence: 'Brooklyn', year: 2});
```

A developer proposes rejecting the second node because it lacks a `residence` property and the third because it contains an additional `year` property.

Which assessment is most accurate?

A. Only the third node is invalid because all nodes sharing a label must have the same number of properties.

B. Only the second node is invalid because every node must contain the properties defined by the first node with the same label.

C. Both nodes are invalid because Neo4j requires property names to be declared in the label before nodes can be created.

D. **Correct:** Both nodes are valid because labels do not require identical property sets, unless additional constraints or application rules impose such requirements. Correct answer

24 MULTIPLE CHOICE CORRECT 1 / 1

Marlize is a student completing the Movie Graph homework. Marlize wants to find people connected to the movie Cloud Atlas and display both the person's name and the type of the relationship found.
Which one of the following queries can Marlize execute to achieve this?

A. **Correct:**
```
MATCH (p:Person)-[r]-(:Movie {title:
"Cloud Atlas"})
RETURN p.name, type(r);
```
Correct answer

B.
```
MATCH (p:Person), (m:Movie {title: "Cloud Atlas"})
RETURN p.name, type(m);
```

C.
```
MATCH (p:Person {title: "Cloud Atlas"})
RETURN p.name, type(p);
```

D.
```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
RETURN p.name;
```

25 MULTIPLE CHOICE CORRECT 1 / 1

A university models student participation in study groups using Neo4j. Each `Student` node may participate in several groups through directed `MEMBER_OF` relationships.
The following query is executed:

```
MATCH (s:Student)-[:MEMBER_OF]->(g:StudyGroup)
RETURN g.name, count(s) AS members
ORDER BY members DESC;
```

Which change would be necessary if the university wanted to display only study groups with more than five members, while retaining the member count in the output?

A. Add `HAVING count(s) > 5` before `RETURN`.

B. **Correct:** Add `WHERE count(s) > 5` immediately before `RETURN`. Correct answer

C. Add `WHERE members > 5` immediately after `ORDER BY`.

D. Use `MATCH` to restrict the groups based on their member count before the aggregation is calculated.

ASSESSMENT CONTENT

26 **FILL IN THE BLANK** INCORRECT 0 / 4

A university's XML database stores research projects. Each PROJECT has a department attribute and contains one or more RESEARCHER elements. Each researcher has a PUBLICATIONS child element containing the number of publications they have produced. The university wants to identify departments where at least one researcher has published more than 5 papers. The result must contain each qualifying department code only once.
Complete the XQuery. Each blank is worth 1 mark.

```
for $D in distinct-values(//PROJECT/@department)
let $R := //PROJECT[ ✕ /@department ]/RESEARCHER
where [ ✕ let ] $P in $R/PUBLICATIONS
      [ ✕ and ] (xs:integer($P) > 5)
return <DEPARTMENT>{[ ✕ $R ]}</DEPARTMENT>
```

Incorrect

Acceptable answers for **BLANK-1** must match this pattern: ^\s*(\[\s*@department\s*=\s*\$D\s*\]|\[\s*\$D\s*=\s*@department\s*\])\s*$

Acceptable answer for **BLANK-2** is **some**
Responses must match exactly

Acceptable answer for **BLANK-3** is **satisfies**
Responses must match exactly

Acceptable answer for **BLANK-4** is **$D**
Responses must match exactly

27 **FILL IN THE BLANK** PARTIALLY CORRECT 1 / 3

SAPS forensics lab stores evidence records in BaseX. Consider the following XML document:

```
<EVIDENCE>
    <ITEM caseId="DF101" type="Disk">
        <HASH verified="true">abc123</HASH>
        <HASH verified="false">def456</HASH>
    </ITEM>
    <ITEM caseId="DF102" type="Phone">
        <HASH verified="false">ghi789</HASH>
    </ITEM>
    <ITEM caseId="DF103" type="Disk">
        <HASH verified="true">jkl012</HASH>
    </ITEM>
</EVIDENCE>
```

(a) What XPath symbol is used to select all descendant elements at any depth from the current context node? ✓ **//**

(b) Complete the predicate using one word to select ITEM elements whose type attribute is Disk.

```
ITEM{ ✗ Blank 2 = 'Disk'} . Note, the curly brackets {} denote square brackets.
```

(c) Complete the XPath expression to retrieve the caseId of the second ITEM containing at least one verified hash, in document order.

```
(//ITEM{HASH{@verified = 'true'}}) ✗ Blank 3 /@caseId
Note, the curly brackets {} denote square brackets
```

Partial Credit

Acceptable answer for **BLANK-1** is **//**
Responses must match exactly

Acceptable answers for **BLANK-2** are **@type**, **type** and **@ type**
Responses must match exactly

Acceptable answers for **BLANK-3** are **2** and **[2]**
Responses must match exactly

28 **FILL IN THE BLANK** PARTIALLY CORRECT 1 / 3

(a) A distributed emergency-response database has three data centres. During a network partition, the system rejects some write requests to prevent conflicting versions of incident records from being accepted.
Which CAP property is the system potentially sacrificing? ✓ **availability**

(b) In the CAP theorem, what property describes a distributed system's ability to continue operating despite communication failures between nodes? ✗ **availability**

(c) During a network partition, a distributed database continues accepting requests at both isolated data centres. However, each centre may temporarily return a different version of the same incident record. Which CAP property is being prioritised over strong consistency? ✗ **consistency**

Partial Credit

Acceptable answer for **BLANK-1** is **Availability**
Responses must match exactly

Acceptable answer for **BLANK-2** is **Partition tolerance**
Responses must match exactly

Acceptable answer for **BLANK-3** is **Availability**
Responses must match exactly

29 **FILL IN THE BLANK** **PARTIALLY CORRECT** 2 / 3

An online travel agency stores booking documents in a MongoDB collection named bookings. Each document contains bookingId, destination, price, status and travellers. Assume the following documents exist:

bookingId	destination	price	status
B101	Cape Town	2500	Confirmed
B102	Durban	1800	Confirmed
B103	Johannesburg	3200	Pending
B104	Gqeberha	2800	Confirmed

(a)

(a) A developer executes the following query:

```
db.bookings.find({
    price: { $gt: 2000 },
    status: "Confirmed"
})
```

How many documents will the query return? ✓ **2**

(b) A developer executes the following query:

```
db.bookings.find(
    {},
    { destination: 1, price: 1, _id: 0 }
)
```

How many fields will be displayed in each returned document? ✓ **2**

(c) A developer executes the following query:

```
db.bookings.find()
    .sort({ price: -1 })
    .limit(2)
```

Which booking ID appears first in the result? ✗ **B102**

Partial Credit

Acceptable answers for **BLANK-1** are **2** and **two**
Responses must match exactly

Acceptable answers for **BLANK-2** are **2** and **two**
Responses must match exactly

Acceptable answer for **BLANK-3** is **B103**
Responses must match exactly

30 **FILL IN THE BLANK** CORRECT 3 / 3

A smart-campus system stores sensor readings in a MongoDB collection named readings. Each document contains a sensorType and a numeric duration, representing the number of seconds a sensor was active.
(a) The administrator wants to group readings by sensor type. Which aggregation stage is used to group documents? ✓ **$group**
(b) Within a $group stage, which accumulator calculates the total of the duration values in each group? ✓ **$sum**
(c)After calculating each group's totalDuration, the administrator wants to retain only groups whose total exceeds 10,000 seconds.
Complete the condition:

```
{
    $match: {
        totalDuration: { ✓ $gt: 10000 }
    }
}
```

Correct

Acceptable answers for **BLANK-1** are **$group** and **group**
Responses must match exactly

Acceptable answers for **BLANK-2** are **$sum** and **sum**

Responses must match exactly

Acceptable answers for **BLANK-3** must match this pattern: ^\s*\$?\s*gt\s*:\s*10000\s*$

31 **FILL IN THE BLANK** CORRECT 3 / 3

An online bookshop stores book descriptions and customer reviews in a MongoDB collection named books. The searchable text fields vary between documents, so the developer wants to create a wildcard text index and search for books containing the phrase climate change.

(a) Complete the index definition by supplying the appropriate index type.

```
db.books.createIndex({ "$**":"✓ text" })
```

(b) Complete the MongoDB text-search operator to search for the phrase climate change.

```
db.books.find({
   $text: { ✓ $search: "climate change" }
})
```

(c) A Java application retrieves the matching books using a MongoDB cursor. Which method checks whether another document is available before the application retrieves it using cursor.next()?

```
while (cursor.✓ hasNext()) {
    Document book = cursor.next();
}
```

Correct

Acceptable answer for **BLANK-1** is **text**

Responses must match exactly

Acceptable answers for **BLANK-2** must match this pattern: ^\s*[$]?\s*search\s*:\s*"\s*climate\s+change\s*"\s*$

Acceptable answers for **BLANK-3** are **hasNext** and **hasNext()**

Responses must match exactly

32 **FILL IN THE BLANK** CORRECT 3 / 3

A vehicle rental company uses Neo4j to model its fleet. The graph contains:

- Vehicle nodes with a registration property.
- Customer nodes with a `customerId` property.
- `RENTED_BY` relationships from vehicles to customers.
- `BOOKED` relationships from customers to rental reservations (`Reservation`).

(a) Complete the Cypher pattern to match vehicles rented by customers. *Note, the curly brackets {} denote square brackets.*

```
MATCH (v:Vehicle)-{✓ :RENTED_BY}->(c:Customer)
RETURN v.registration
```

(b) Complete the relationship type to find rental reservations booked by a customer. The relationship points from the customer to the reservation. *Note, the curly brackets {} denote square brackets.*

```
MATCH (c:Customer)-{✓ :BOOKED}->(✓ r:Reservation)
RETURN r
```

Correct

Acceptable answers for **BLANK-1** are **:RENTED_BY** and **: RENTED_BY**
Responses must match exactly

Acceptable answers for **BLANK-2** are **:BOOKED** and **: BOOKED**
Responses must match exactly

Acceptable answers for **BLANK-3** are **r:Reservation** and **r: Reservation**
Responses must match exactly

33 **FILL IN THE BLANK** **PARTIALLY CORRECT** 2 / 3

A South African wildlife conservation organisation uses Neo4j to store information about rangers and the province in which they operate.
Assume the following Ranger nodes exist:

rangerId	name	province	experience
R01	Lerato	Limpopo	8
R02	Thabo	Mpumalanga	4
R03	Pholosho	Limpopo	6
R04	Mpho	KwaZulu-Natal	3

Consider the following Cypher query:

```
MATCH (r:Ranger)
WHERE r.experience >= 5
RETURN DISTINCT r.province
ORDER BY r.province
```

(a) How many rows will the query return? ✓ **1**

(b) Which province is/are returned? ✓ **Limpopo**

(c) Consider the following Cypher query:

```
MATCH (r:Ranger)
WHERE r.experience >= 4
```

```
RETURN r.name
ORDER BY r.experience DESC
```

What is the name returned in the last row of the result? ✕ **Pholosho**

Partial Credit

Acceptable answers for **BLANK-1** are **1** and **one**
Responses must match exactly

Acceptable answer for **BLANK-2** is **Limpopo**
Responses must match exactly

Acceptable answer for **BLANK-3** is **Thabo**
Responses must match exactly

ASSESSMENT CONTENT

34 **ESSAY** **PARTIALLY CORRECT** 14 / 20

Study the following scenario and answer the questions that follow:

A Gauteng emergency medical services provider operates ambulances across Johannesburg, Pretoria and surrounding municipalities. The provider uses MongoDB to manage emergency dispatch records.
Each document in the dispatches collection has the following structure:

```
{
    dispatchId: "D501",
    station: "Pretoria",
    status: "Completed",
    priority: "High",
    responseTime: 11,
    patient: {
        age: 64,
        municipality: "Tshwane"
    },
    crew: [
        {name: "Thabo", role: "Paramedic"},
        {name: "Lerato", role: "Driver"}
    ],
    treatments: [
        {code: "OXY", cost: 250},
        {code: "ECG", cost: 400}
    ],
    keywords: ["cardiac", "urgent", "adult"]
}
```

The database contains thousands of dispatch records from several emergency stations. Management wants to identify slow responses, investigate treatment costs and analyse dispatch performance.

Source:Makura.S.M (2026)

(a) Write a MongoDB query to retrieve the dispatchId, station and responseTime of dispatches satisfying all the following conditions:

- The dispatch status is "Completed".
- The response time exceeds 10 minutes.
- The crew includes a member whose role is "Paramedic".
- The keywords array contains "cardiac".

Your query must return only the specified fields, excluding the default _id field. (5 Marks)
(b) Following a policy change, every dispatch with priority "High" must have the keyword "review-required" added to its keywords array. The keyword must not be duplicated if it already exists.
Write a MongoDB update operation that applies this change to all matching documents. (4 Marks)
(c) Management wants to compare completed dispatches across stations. Write an aggregation pipeline that:

1. Selects only dispatches with status "Completed". (2)
2. Groups the matching documents by station and calculates the average response time and number of dispatches for each station. (3)

3. Sorts the results by average response time in descending order. (2)

(d) Suppose the completed dispatches from one station have response times of 8, 12, 15 and 25 minutes.

1. Calculate the average response time returned by the pipeline. (2)
2. If the station's response time average is considered acceptable only when it is at most 15 minutes, state whether this station meets the target. (2)

Total: 20 marks

Answer

a) db.dispatches.find({
  status: "completed",
  responseTime: {$gt: {10}},
  crew: {elemMatch: {"role" : "Paramedic"}},
  keywords: ["cardiac"]
})

b) db.dispatch.updateMany({priority: "High},$append:{keywords: "review-required"})

c) db.dispatch.aggregate([
  {$match:{status: "Completed"}},
  {$group: {_id: "station", avgResponseTime: {$avg: {"responseTime"}}, $sum:{1}}},
  {$sort: {avgResponseTime: -1}}
])

d) 1) 15
2) Yes it meets the target

(d has really confused me because theres no real indication that we should write code and specific values are given which makes it feel like something to calculate, if I have completely misunderstood this question I apologise and hope you get a good laugh out of my silly response)

Example of a correct response

Answer:
(a) MongoDB Query [5 marks]
Answer:

```
db.dispatches.find(
    {
        status: "Completed",
        responseTime: { $gt: 10 },
        "crew.role": "Paramedic",
        keywords: "cardiac"
    },
    {
        _id: 0,
        dispatchId: 1,
        station: 1,
        responseTime: 1
    }
)
```

Marking rubric:

Criterion	Marks
Correctly filters status as "Completed"	1
Correctly filters response time greater than 10	1
Correctly queries the nested crew role	1
Correctly checks that keywords contains "cardiac"	1
Correctly projects the three required fields and excludes _id	1
Total	**5**

(b) MongoDB Update [4 marks]

Answer:

```
db.dispatches.updateMany(
    { priority: "High" },
    {
        $addToSet: {
            keywords: "review-required"
        }
    }
)
```

Marking rubric:

Criterion	Marks
Uses updateMany() to update all matching documents	1
Correctly selects documents with priority "High"	1
Uses $addToSet to add the keyword without duplicating it	1
Correctly specifies the keywords array and "review-required" value	1
Total	**4**

Important: $addToSet adds the keyword only if it is not already present in the array. Unlike $push, it prevents duplicate values.

(c) Aggregation pipeline [7 marks]

Answer:

```
db.dispatches.aggregate([
    {
        $match: {
            status: "Completed"
        }
    },
    {
```